

**БУЛ БИО - НЦЗПБ ЕООД****BB - NCIPD Ltd.**

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**CERTIFICATE of ANALYSIS**

**BCG Vaccine, Freeze-dried**  
powder for suspension for injection - 20 ampoules x 10/20 doses  
**Lot № 728-2**

Date of manufacture: May 2022

| №   | Test / Method                                                                    | Acceptance Criteria                                                                                                                                                                                                                                                                                                                    | Result                 |
|-----|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1.  | Appearance                                                                       | Dry mass powder. After reconstitution - homogenous, slightly opalescent suspension                                                                                                                                                                                                                                                     | Pass                   |
| 2.  | Colour                                                                           | Dry whitish mass powder                                                                                                                                                                                                                                                                                                                | Pass                   |
| 3.  | Identification                                                                   | Ziehl-Neelsen method - microscopic examination of the bacilli in stained smears, demonstrating their acid-fast property (single violet-red straight or slightly curved rods).<br>Characteristic appearance (dry, raised, rough surface, uneven periphery with pale yellow hue) of the colonies grown on solid medium Lowenstein-Jensen | Pass                   |
| 4.  | Sterility                                                                        | Absence of microbial growth (except for the presence of mycobacteria)                                                                                                                                                                                                                                                                  | Sterile                |
| 5.  | Virulent mycobacteria                                                            | Absence of virulent mycobacteria                                                                                                                                                                                                                                                                                                       | Pass                   |
| 6.  | Total bacterial content (opacity at $\lambda=490$ nm)                            | Extinction: $0.380 \pm 0.060$                                                                                                                                                                                                                                                                                                          | 0.390                  |
| 7.  | Count of viable units (NVU) on Lowenstein-Jensen medium (last satisfactory test) | $1.5 \times 10^6 - 6.0 \times 10^6$ NVU/ml                                                                                                                                                                                                                                                                                             | $2.9 \times 10^6$ / ml |
| 8.  | Thermal stability                                                                | The NVU in the heated vaccine is not less than 20 per cent of that in the unheated vaccine                                                                                                                                                                                                                                             | 24.14 %                |
| 9.  | Vacuum test                                                                      | Presence of vacuum                                                                                                                                                                                                                                                                                                                     | Pass                   |
| 10. | Loss on drying                                                                   | Not more than 3.0 per cent                                                                                                                                                                                                                                                                                                             | 0.011 %                |
| 11. | Storage conditions                                                               | 2 °C to 8 °C: protected from direct sunlight                                                                                                                                                                                                                                                                                           |                        |
| 12. | Expiry date                                                                      | 36 months from the date of the last satisfactory test for determination the number of viable units                                                                                                                                                                                                                                     | May 2025               |

**Conclusion:**

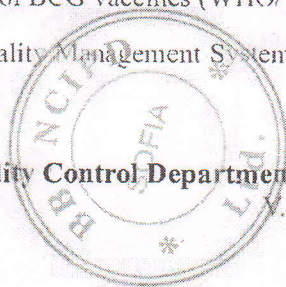
BCG Vaccine, freeze-dried lot № 728-2 meets all national requirements and satisfies WHO recommendations to assure the quality, safety and efficacy of BCG vaccines (WHO/TRS 979) and European Pharmacopoeia.

The Vaccine has been produced in a firm with approved Quality Management System Standards: ISO9001:2015 and ISO 14001:2016.

Date of issue: 13.09.2022

Head of Quality Control Department:

V. Milanova, PhD





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## CERTIFICATE OF ANALYSIS

**Diluent (Saline) for BCG Vaccine**  
Sol.inj., amp. – 1.0 ml, x 20

**Lot № 518-1**

Date of manufacture: July 2022

| №  | Test/Method                               | Acceptance Criteria                                                               | Result    |
|----|-------------------------------------------|-----------------------------------------------------------------------------------|-----------|
| 1. | Appearance                                | Colourless transparent liquid                                                     | Pass      |
| 2. | Extractable<br>volume<br>Ph. Eur. 2.9.17. | Not less than 1.0 ml                                                              | Pass      |
| 3. | pH<br>Ph. Eur. 2.2.3.                     | 5.00 – 7.00                                                                       | 6.78      |
| 4. | Sodium Chloride<br>Ph. Eur. 2.2.20.       | 0.65 – 1.15 per cent                                                              | 0.88 %    |
| 5. | Sterility<br>Ph. Eur. 2.6.1.              | Sterile                                                                           | Pass      |
| 6. | Pyrogens<br>Ph. Eur. 2.6.8.               | Not more than 1.15 °C<br>(if summed response of three rabbits<br>does not exceed) | 0.1 °C    |
| 7. | Storage conditions                        | Store at temperature below 25 °C<br>Do not freeze!                                |           |
| 8. | Expiry date                               | 60 months                                                                         | June 2027 |

### Conclusion:

**Diluent (Saline) for BCG Vaccine Lot № 518-1** meets all European Pharmacopoeia requirements. The product has been produced in a firm with approved Quality Management System Standards: ISO 9001:2015 and ISO 14001:2016

Date of issue: 13.09.2022

Head of Quality Control Department:

V. Milanova

